

ChatGPT in Higher Education: Perceptions of Computer Science-Related Students

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Abstract—In the field of education ChatGPT has sparked both admiration and controversy. This study explores students' perspectives, specifically focusing on those in computer science-related fields. We investigated their motivations, trust, perceptions of utility, and reliability of ChatGPT by conducting two surveys—one at the beginning and another at the end of the semester. During the semester, students were encouraged to engage with ChatGPT. Our findings highlight the tool's multifaceted use in an academic settings, establishing it as a valuable resource for a variety of learning tasks. Most students have incorporated ChatGPT into their regular academic activities and view it as a beneficial aid. They perceive it as a multi-task solver and anticipate significant advancements in its writing assistance features in the near future. Many, not only attribute high accuracy to it but think that it adheres to appropriate content and structural norms. Our results suggest that active confrontation with ChatGPT enhances understanding of its capabilities, limitations and autoregressive nature. Consequently, we recommend an approach of informed engagement that includes the distinction between language processing and genuine language understanding and a carefully crafted terminology.

Index Terms—ChatGPT, large language models, higher education, educational technology

I. INTRODUCTION

In recent years, researchers made significant advancements in natural language processing (NLP) with no sign of slowing down. Consequently, large language models (LLMs), such as ChatGPT, entered the realm of education which elicited both admiration and controversy [1]. The convenience of accessing a seemingly knowledgeable artificial communication partner with a few mouse clicks anytime and anywhere seems like a blessing for anyone who wants to learn, especially when one considers that personalized feedback is a cornerstone of effective learning [2]. A constructivist learning experience with personalized tutoring, automated essay grading and both, interactive and adaptive learning modalities is potentially within reach [3], [4]. Conversely, many fear the loss of important student skills in combination with a high dependency on powerful but also proprietary and opaque tools. A predominant concern aligns with the potential undermining of academic

integrity [5] and the end of online exams [6]. Excitement and skepticism, including ethical concerns, were expressed by many different interest groups. This study however, focuses on how students perceive and deal with ChatGPT (based on GPT-3.5) and which benefits and challenges they face.

For university students, chatbots can assist in research and writing tasks [7], and they can foster the cultivation of critical thinking and problem-solving skills [5]. Specifically for computer science students, large language models can function as collaborative partner in software development. Moreover, ChatGPT is not only able to generate and summarize text, but it can also improve the readability, clarity and quality of written work. It is equipped to craft language-specific summaries and comprehension exercises, which can help students and educators improve their language skills. ChatGPT as well as LLMs trained or fine-tuned on code are also capable of generating and explaining code—a potential challenge for educators, yet a valuable tool that can expedite students' learning progress.

Autoregressive self-supervised models, such as GPT-3 [8], GPT-4 [9], LLaMA [10], Megatron-Turing NLG [11], which are based on the transformer architecture [12], are trained on massive amount of textual data to predict the next token within a sequence of tokens. This approach to generate text one word at a time has led to the unexpected effect of models learning multiple tasks without any explicit supervision [13]. Increasing the number of model parameters, training data and duration, does enhance the model's ability to perform a broader range of tasks. Examples include question answering, machine translation, reading comprehension, summarization, coding, text-based games, music generation, and even solving mathematical problems [14].

While early chatbots relied on pattern matching and string processing, contemporary chatbots powered by artificial intelligence (AI), or more precisely deep learning (DL), harness high-dimensional probability distributions, learned by observations and aligned by human feedback. Chatbots, like ChatGPT, are fine-tuned and aligned large language models tailored for interactive dialogues with users. By ingesting past interactions

as context, these chatbots can create an illusion of ‘recalling’ previous conversations. Their learning mechanism is anchored in a reward model, predominantly trained via human feedback (HF). Such feedback steers large language models to learn specific policies through reinforcement learning (RL). Human feedback is required because, in contrast to a game such as Go or chess where the quality of a move can be computed, the rewards to give a valuable response are often not easy to define or measure, especially when dealing with complex tasks that involve human values or preferences. Human feedback pushes LLMs to generate human-like answers.

However, as argued by Bender and Koller [15], one should not mistake the appearance of human-like understanding, i.e. NLP, as natural language understanding (NLU). By confusing NLP with NLU the “seemingly coherent language [generated by LLMs] heightens the potential for automation bias, deliberate misuse, and amplification of a hegemonic worldview” [16]. Since ChatGPT was diagnosed to be politically biased [17], [18] and its opinionated responses can affect what users write and think [19], utilizing it prompts a challenge for students and educators alike. Furthermore, even if the model improvement in accuracy in relation to resource expansion is predictable, specific model capabilities, inputs, and outputs cannot necessarily be predicted ahead of time [20].

While the capabilities of large language models to generate human-like text and the capability of chatbots to take part in a human-like conversation is impressive, they are not able to simulate the same level of understanding and empathy a human teacher or professor possesses. Undoubtedly, ChatGPT can be used to solve a broad variety of tasks, however, it is not specifically tailored towards learners. For example, it does not automatically explain each step in a chain of thought. This can be mitigated by providing additional context to the prompt.

Furthermore, unlike databases or search engines, LLMs lack the ability to cite their sources accurately, as they generate text through extrapolation from the provided prompt—their output is contingent. This extrapolation can result in hallucinated outputs that do not align with the training data. As observed by [21], the limited contextual understanding of LLMs—as they abstract information from the prompt and training data—can contribute to hallucination by potentially overlooking crucial details. Consequently, users who rely on the current generation of LLMs must be aware of the possibility of false, fabricated and biased information and take necessary precautions to mitigate the associated consequences. Students have to be suspicious and have to double check if the provided answers are correct.

Overall, ChatGPT is a simulation of a knowledgeable but also biased and untrustworthy partner with certain blind spots which is capable of generating multiple suggestions to a broad range of requests. To learn effectively with it, students require the ability to spot biases and errors, which is only possible if they already have a certain level of understanding, metacognitive awareness and the willingness to reflect.

In the subsequent section, we provide a short review of existing literature that explores the efficacy of ChatGPT in

addressing exam questions, alongside an examination of the students’ interactions with, and perceptions of, this emergent technological tool. Following this, we delineate our research objectives, explicate the experimental framework, and elucidate the methodology employed in our study. The fourth section presents our empirical findings, derived from a survey administered to 81 participants, coupled with a secondary evaluation of a subset comprising 39 students conducted at the semester’s conclusion. This dual-phase assessment facilitates a comparative analysis aimed at discerning variations before and after the course. We conclude with a discussion of the implications for educators in this evolving educational paradigm by advocating for an informed engagement.

II. RELATED WORKS

Numerous studies have delved into the capabilities of ChatGPT from an educational vantage point. Within the domain of medical education Gilson et al. [22] assessed ChatGPT’s performance on questions from the United States Medical Licensing Examination Step 1 and Step 2. Their findings bolster the potential of ChatGPT as a collaborator in medical education. This sentiment is echoed by Kung et al. [23], as ChatGPT’s scores hovered near or surpassed the passing benchmark. However, a contrasting study by Huh [24] juxtaposed the performance of medical students in Korea with that of ChatGPT on a parasitology exam and determined that the latter lagged behind in both knowledge and interpretative capacity. Drori et al. [25] report that the largest GPT-3 model solves 18.8% of university mathematical questions using zero-shot learning and 30.8% using few-shot learning. Performance surged to 81% when utilizing Codex, an LLM fine-tuned for coding. Yet, this study faced criticism from Davis [26], who took issue with attributing the achievement to “a neural network” rather than the Python package SymPy [27]. Additionally, he dismissed a line-by-line code explanation as an insufficient rationale.

Mirroring the objectives of our research, Shoufan [28] explored students’ experience with ChatGPT and their overarching perception. Initially, they posed an open-ended question to students. Following a qualitative textual analysis, 8 positive and 7 negative themes were identified. A subsequent questionnaire, formulated based on these themes, was administered post three learning activities involving ChatGPT. A key insight was that students believed adequate background knowledge in the relevant study area was essential for critically assessing ChatGPT’s responses. Despite its moderate answering accuracy, the study revealed that ChatGPT seems to be an attractive platform for students. In [29] Ali et al. investigated ChatGPT’s impacts on English language acquisition. They argue that the chatbot generally motivates learners to develop their reading and writing skills. Von Garrel et al. [30] conducted a quantitative survey involving 6300 students in Germany. Their findings suggest a high usage of ChatGPT (48.9%) among students with DeepL (12.3%) at the second place. Furthermore, they posit that the usage of AI-based tools is diverse but

research (45.4%), translations (42.2%) and explanations of subject specific concepts (56.5%) are among the most relevant.

Tholander and Jonsson [31] conducted a quantitative study examining the role of large language models and generative AI in enhancing creative design and ideation. They emphasize the contrast in communication capabilities between natural language interfaces and conventional graphical user interfaces. Users often import expectations from their experiences with natural language interactions, particularly from conversations with other individuals, and modify their responses based on these expectations. This underscores the crucial need to differentiate between natural language processing (NLP) and natural language understanding (NLU). Tholander and Jonsson observed that while generative AI facilitates rapid generation of design alternatives, it falls short in depth of understanding, offering only limited memory of ongoing interactions due to its lack of contextual comprehension. Similarly, Chiou et al. [32] conducted a focused case study on interactions with image generation AI, revealing the technology's capacity to expand human creativity and to introduce unique perspectives in artistic expressions. Through a *research through design (RTD)* approach, they explored collaborative co-creation in design. Their findings indicate that image generators promote divergent thinking and ideation within the design process, serving as valuable tools for creative exploration. The study observed that abstract prompts tended to yield more varied outcomes, whereas generating specific images necessitated detailed and precise descriptions. This distinction highlights the nuanced capabilities and limitations of AI in the creative domain.

III. GOALS

This study aims to investigate several key aspects concerning the integration and utilization of AI, particularly ChatGPT, among students enrolled in computer science-related programs. Our first objective is to discern whether the contemporary education system can expect a generation of students who are increasingly relying on AI systems like ChatGPT to complete a significant portion of their academic tasks. Through this lens, we aim to elucidate ChatGPT's burgeoning role in the educational landscape.

Additionally, our research delves into the specific usage patterns of ChatGPT among computer science-related students, exploring their motivations, concerns, and trust levels in utilizing such AI tools for various academic purposes. We seek to comprehend the diversity of roles that ChatGPT fulfills within the realm of their studies and how students perceive the utility and reliability of these digital resources.

Furthermore, this study endeavors to evaluate the impact of incorporating AI, like ChatGPT, into the learning process of Computer Science and Design students, with a particular focus on design processes and prototype development. We aim to discern the extent to which these students leverage AI in their creative and problem-solving endeavors, and how they perceive the benefits and challenges of doing so. By exploring these facets, we hope to contribute to a deeper understanding

of the multifaceted relationship between generative AI such as ChatGPT and academic education.

IV. METHOD

In our study, a cohort of 81 students enrolled in computer science-related programs participated. They were at least in the second semester of their study-program. We developed a survey which was conducted at the onset of the summer semester in 2023. Additionally, we conducted a longitudinal assessment of a subset of students from the Computer Science and Design program throughout the term, totaling 39 participants.

A. Instructional setting of the subsample

The 39 participants of the course were bachelor students affiliated with the department of the Munich Center for Digital Sciences and AI (MUC.DAI). In terms of the sample characteristics, the students participating in this study possess a background in computational thinking, proficiency in the programming language Python, and have completed foundational coursework in their first semester, covering topics such as interface and interaction design, as well as fundamental principles in design and typography. They were enrolled in the second semester course Project Module: Processes in the study program Computer Science and Design. The objective of the course was to conceptualize and prototype a new generative AI application that could be a benefit to society. This involved research about current generative AI technology and use cases, joint ideation and conceptualization of a generative AI product and its execution in the form of a prototype. Across all steps, integrating generative AI as partner for the working groups was encouraged. Upon completing the course, students are equipped to classify and identify design, innovation and (agile) software development processes and apply them independently as required. In a project-based setup, they are introduced to agile methodologies and digital tools that span both computer science and design domains. This setup facilitates their understanding of the application of previously acquired competencies in programming languages, design methodologies, and domain-specific tools. Moreover, students gain the capacity to introspect on their professional journey, critically evaluate their methodologies, and contribute uniquely to their teams, all while contemplating real-world challenges.

The student teams, comprising four to six members with varied skills, were required to integrate generative AI into their projects to explore its impact on their professional fields. This integration involved assigning a rotating team role to a virtual colleague, exemplified by ChatGPT this year. After each project deliverable, students critically reflected on the benefits, challenges, and ethical considerations of this approach in written form. Additionally, to successfully complete the course, teams were obligated to submit various work deliverables, presenting and defending their project milestones before their peers.

B. Questionnaire

We designed a questionnaire to evaluate students' engagements with ChatGPT. This tool was administered at the start of the semester, and again at its conclusion for the subset of students. The end-of-semester version of the questionnaire included additional questions compared to its initial counterpart.

This questionnaire was structured to cover various topics, including students' self-assessment of competencies relevant to the course, their usage patterns and challenges with ChatGPT, and attitudes toward the tool. It also explored the perceived impact of ChatGPT on learning and work processes, assessed students' understanding of how ChatGPT functions, and gathered their perspectives on the tool's future development and its potential influence on their lives. An overview with sample items is provided in Fig. 1.

Unique to the end-of-semester survey was a "prompt" section where students were tasked with crafting a single prompt for ChatGPT to address a specific problem. This open-ended query aimed to shed light on students' capability to draft an effective prompt.

Topics	Example items
Self-assessment of competences	I feel confident in the use of computers. I am able to go through a software development process independently. I am very interested in programming.
Use of ChatGPT	How often do you currently use ChatGPT? How often did you use ChatGPT in the course? In what ways did you use ChatGPT during the course?
Attitude towards ChatGPT	What are the current limitations of ChatGPT? Select all aspects that describe the current limitations correctly. Do you want to work with ChatGPT? How much do you trust the answers from ChatGPT? My impression is that I developed fewer skills using ChatGPT than I would have without using it. How important is it to you to be able to understand how ChatGPT reaches its conclusion?
Perceived influence of ChatGPT on different facets of learning- and working-process	In what areas of your day-to-day work do you think generative artificial intelligence is helpful to you now/in 5 years? Now, at the end of the semester, how do you evaluate the impact of ChatGPT on the following points of the course (learning success, errors in the product, creativity of the work process, diversity of the results ...)?
Knowledge about functionality of ChatGPT	Is ChatGPT likely to provide you with neutral or empirically tested facts (value judgment free) or normative (i.e. morally evaluated) answers? Which task does ChatGPT solve? Choose the answer that suits you best. (Try to provide correct information to my queries; Try to generate a text that convinces me; ...)
Forward-looking thoughts about development and influence of ChatGPT	I see my own professional opportunities threatened by AI-based programming assistants or other generative AI technologies. With the help of AI-based programming assistants or other generative AI technologies, even non-technical users will be able to develop software/web applications in the future. AI-based programming assistants or other generative AI technologies will lead to a constant change in the requirements and competence profile of computer scientists.

Fig. 1. Different topics of the questionnaire

The questionnaire items employed a Likert scale response

format (ranging from 1 - strongly agree to 5 - not agree). Additionally, the survey incorporated three open-ended questions.

V. ANALYSIS AND RESULTS

In the following section we unfold our findings derived from an examination of student engagement with ChatGPT. Initially, we present a general analysis based on the collective responses from the entire sample, revealing insights into the frequency of ChatGPT usage for academic purposes, alongside students' perceived value and future expectation of this tool. Subsequently, we explore specific aspects, such as the types of usage within a course setting, challenges encountered, and the nuanced skill of prompt engineering. Additionally, this section includes an analysis on a subsample, offering a comparative perspective on changes in usage patterns and attitudes towards ChatGPT from the beginning to the end of the semester.

A. Results for the whole sample

A noteworthy 54% of students reported utilizing ChatGPT for academic purposes at least once a week. In contrast, a mere 13% reported never having used ChatGPT, a figure considerably lower than the non-usage rate for any AI tool, as documented by Garrel et al. [30]. This regularity in usage signifies its active role in their academic routines. 11% of students disclosed that they also use other generative AI tools at least once a week or more, demonstrating a growing familiarity with and reliance on such technologies.

1) Students' perceived value and future prospects of ChatGPT: The study's findings show that participants currently view ChatGPT as a valuable resource in the areas of text-writing ($M = 1.64$, $SD = .74$), communication ($M = 2.05$, $SD = 1.11$), knowledge-research ($M = 2.01$, $SD = .90$), and individual learning ($M = 1.95$, $SD = .89$). The greatest observed difference in perceived usefulness is in the realm of writing assistance, followed by its application in learning support, knowledge research, and communication.

Study participants hold strong expectations regarding the evolving role of ChatGPT in academic settings over the next five years. They firmly believe that ChatGPT will increasingly serve as an invaluable writing assistant in the future ($M = 1.37$, $SD = .51$, $V = 586.5$, $p = 3.003e-05$) compared to its current capabilities. Their vision includes ChatGPT evolving to provide more advanced and nuanced support, facilitating the creation of documents with enhanced ease and precision.

In the realm of knowledge research, participants anticipate ChatGPT refining its capabilities ($M = 1.37$, $SD = .70$, $V = 1130$, $p = 5.259e-08$). They also foresee ChatGPT excelling as a communication aid ($M = 1.53$, $SD = .79$, $V = 890.5$, $p = 0.0004$) beyond its current functionality. Additionally, participants express the belief that ChatGPT will become more effective as a learning tool ($M = 1.33$, $SD = .71$, $V = 915$, $p = 2.984e-07$) compared to its present utility.

2) Students' knowledge about ChatGPT: In response to the query regarding the primary task that ChatGPT aims to achieve, the analysis of student responses revealed an intriguing distribution. Notably, 43% of participants selected

the response indicating that ChatGPT endeavors to deliver accurate information in response to their queries. This result underscores the prominent role of information accuracy in the users' expectations from ChatGPT. In contrast, a smaller segment of the respondents, constituting 14%, indicated that ChatGPT's purpose is to generate persuasive text, highlighting the diversity of user perceptions. A similar proportion of 43% of students, in a noteworthy parallel to the first group, believed that ChatGPT's objective is to generate text that not only conveys accuracy but also adheres to appropriate content and structural norms. This outcome underscores the multifaceted nature of ChatGPT's perceived task, emphasizing the need for a nuanced understanding of its role in fulfilling user expectations.

A substantial majority of students (89%) demonstrated awareness that ChatGPT does not limit its capabilities to a narrow range of topics. This finding underscores the prevailing perception among students that ChatGPT possesses a broad and versatile capacity to engage with a wide array of subjects and inquiries.

Furthermore, the survey results revealed that a significant proportion of respondents, accounting for 76%, were cognizant of ChatGPT's limitations stemming from the data on which it was trained. This suggests an understanding among the student population of the inherent constraints in the system's knowledge base. Additionally, a notable 77% of participants indicated that ChatGPT lacks self-awareness regarding the specific texts it employs to generate a given response. This acknowledgment highlights the perception that ChatGPT operates without consciousness of the individual sources of information it draws upon in formulating its outputs.

In this study, a comparison between computer science and design students was omitted due to the heterogeneity in their academic levels, with computer science students from various semesters and design students exclusively from the second semester.

B. Results for the subsample

Analyzing the results of the subsamples gives insights in how students used ChatGPT during the semester and if there was a change in usage and attitude toward the usage.

1) *Students' type of use of ChatGPT in the course:* At the semester's end, students reported on their usage patterns for ChatGPT within the course. Almost all students (97.4%) utilized ChatGPT as *writing assistance*. Over half the students employed it as a *team-partner* (65.8%) and for *research purposes* (73.7%). Approximately one-third leveraged ChatGPT for *support with programming tasks* (36.8%) and *provision of feedback* (28.9%). Less commonly, students used ChatGPT for *control of the result* (23.7%) and as a *communication aid* (13.2%) (see Fig. 2). Through an open-ended question, students also indicated other uses for ChatGPT, such as *brainstorming and inspiration*, and *proof-reading of texts*.

Compared to the start of the semester, it's evident that nearly all students have adopted ChatGPT as a versatile tool for various purposes.

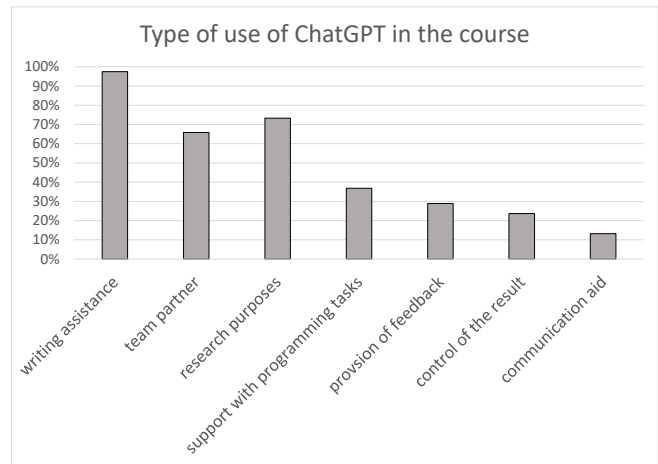


Fig. 2. Type of usage of ChatGPT during the course

2) *Students' challenges in using ChatGPT:* At the semester's conclusion, students were prompted with both closed and open questions regarding the challenges they encountered when utilizing ChatGPT. 71.1% of the students stated that the responses from ChatGPT were of insufficient quality. Nearly half (44.7%) expressed concerns about the accuracy of ChatGPT's answers, noting they were either incorrect or distorted. Additionally, many found assessing the tool's answers challenging. Approximately one-third (31.6%) identified a personal lack of proficiency in effectively communicating with ChatGPT to obtain useful responses.

3) *Students' ability to prompt:* Prompt engineering is becoming a useful skill for facilitating effective communication with large language models (LLMs). Essentially, it involves programming LLMs through specific prompts, as noted by [33]. A prompt, as defined by [34], is a set of instructions given to an LLM [35]. Different prompt patterns exist, as discussed by [33]. These instructions tailor the LLM's behavior, enhancing or refining its capabilities.

In our second survey, students were tasked to craft a single prompt for ChatGPT to address a specific problem:

"You are a group of computer scientists and designers and your group wants to found a new business which promotes the realization of the Sustainable Development Goals (SDGs). Write a prompt to develop ideas for a possible product."

Many students simply rephrased the question as the following:

"Envision being a startup founder with proficiencies in both computer science and design, aiming to establish a company dedicated to realizing the Sustainable Development Goals. What product ideas would you propose for this enterprise?"

Several students appended supplementary details, such as the desired number of ideas, team size, a call for unique and potentially impactful suggestions, or a focus on particular SDG

facets, like social justice. Only two students adapted the so called *Flipped Interaction Pattern* [33]:

“Generate concrete product ideas in the field of software to achieve the Sustainable Development Goals. Ask me as many questions as you need to generate innovative ideas.”

It’s worth noting that, since “SDG” is a recognized term within ChatGPT’s training data, further elaboration on the Sustainable Development Goals did not noticeably enhance its output quality. However, when the question is simply reframed, the LLM’s response tends to be generic. It suggests familiar fields and product ideas, including optimizing crop yields, blockchain systems for trade, virtual reality applications for therapy and education, real-time ecological monitoring through IoT, smart grid technologies, and AI-powered waste management.

Despite the diverse and frequent use of ChatGPT, there still seem to be numerous challenges in dealing with it. These challenges stemmed from both the inherent limitations of the tool and the students’ proficiency in harnessing its capabilities. For educators interested in integrating chatbots powered by large language models (LLMs), we recommend a brief review of different prompt patterns as described in [33]. However, we view prompt engineering not as a specialized field of expertise, but rather as a technique that is easy to learn.

4) *Pre-and-post analyses*: We administered the majority of our questions at both the onset and conclusion of the semester. Our objective was to understand students’ initial perceptions of ChatGPT usage and to compare them with their assessments after gaining extensive experience with the tool throughout the course. We were interested whether these experiences were reflected in different response behaviour.

Fig. 3 illustrates how often students use ChatGPT in general at the beginning of the semester compared to its end. By the semester’s end, 47.4% of students engaged with ChatGPT daily, and another 47.4% utilized it weekly. In comparison, at the beginning of the semester, only 15.4% used it daily, whereas 61.5% used it weekly and 20.5% used ChatGPT monthly or once.

The *t*-test for dependent samples was used to examine which questions showed significant differences in the assessments at the beginning and end of the semester. Of $N = 18$ students, the questionnaires from the beginning and end of the semester were clearly attributable. Compared to the beginning ($M = 2.72$; $SD = .75$), students at the end of the semester ($M = 2.00$; $SD = .77$) reported having more methodological competence in the area of ‘processes’, $t(18) = 4.58$, $p < 0.001$; $d = .67$. According to Cohen [36], this difference is medium. This can be interpreted as a sign for a successful course concerning the goal that students develop special competences.

Moreover, by semester’s end ($M=2.41$, $SD=.51$), students expressed diminished trust in ChatGPT, as compared to its outset ($M=1.94$, $SD = .83$), $t(18) = -2.06$; $p = .028$; $d = .94$. This corresponds to a strong effect. Concurrently, their preference to integrate ChatGPT into their coursework dwindled at the close

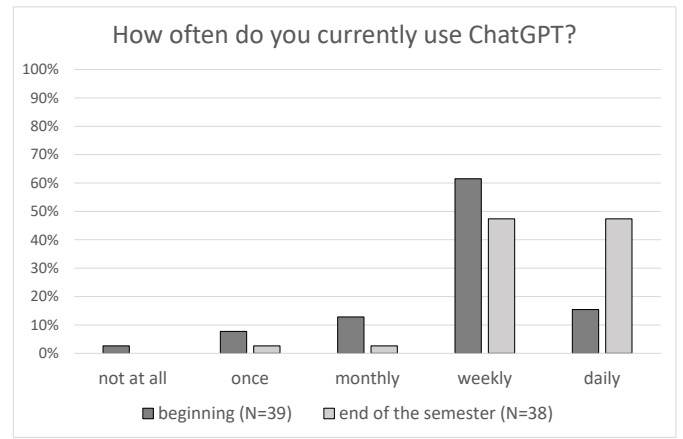


Fig. 3. Usage frequency of ChatGPT at the beginning and end of the semester

($M = 1.94$, $SD = .64$) in relation to the beginning ($M = 1.39$, $SD = .78$), $t(18) = -2.40$; $p = .014$; $d = .98$. These outcomes may reflect an evolving, more discerning stance toward ChatGPT over the semester’s duration.

Nevertheless, by the semester’s conclusion ($M = 1.61$, $SD = .70$), students perceive a higher level of support from ChatGPT in going through the design process than they initially assessed at the beginning ($M = 2.06$, $SD = .87$) ($M = 1.39$, $SD = .78$), $t(18) = 1.81$; $p = .044$; $d = 1.04$. This might suggest that students, throughout the term, improved their ability to effectively utilize ChatGPT to augment their workflow.

VI. DISCUSSION AND OUTLOOK

With the recent presentation of Google’s multimodal LLM called Gemini [37], it seems LLM researchers have achieved another leap, this time towards multimodal conversations and edge computing. Students have to be prepared for utilizing these tools effectively but also to understand generative artificial intelligence to such a degree that they can navigate their environment and contribute to society on their own terms.

Our results underscore the multifaceted utilization of ChatGPT in students’ academic pursuits, positioning it as a versatile instrument for various learning-related tasks. The vast majority of students have readily integrated this tool into their regular academic activities.

Given this widespread adoption, a critical question emerges: How can educators navigate this landscape and equip students to maximize the advantages of this technology while circumventing its potential pitfalls? A significant portion of the students recognized that ChatGPT’s, contingent on its training data, could harbor biases and inaccuracies. Yet, many struggled to articulate the chatbot’s core functionalities accurately. Since complete avoidance is neither realistic nor desirable, we advocate for an informed engagement approach. Such engagement includes the nuances between language processing and genuine language understanding, laying a robust foundation for students’ and educators’ interactions with ChatGPT and other LLM-based tools. Our belief is strengthened by our observation

that students become more suspicious after actively engaging with the chatbot.

A first step could be to actively integrate ChatGPT and other generative AIs into the course. This means to offer tasks of different competence levels, which require the use of such tools, but also to reflect and critically question the use and benefits afterwards. Students should be able to formulate their problem, select the right AI-tools, interact with them critically and reflect on their experience. Different prompts which lead to correct but also to incorrect and inaccurate results had to be discussed. Educators should demystify generative artificial intelligence by provoking errors and effects showcasing limitations such as the so-called reversal curse [38]. They also could and probably should embrace the opportunity to learn from their students. Additionally, fostering precision in terminological usage when discussing large language models is important, particularly to avoid the confusion with anthropomorphic descriptions.

Consequently, there are new starting conditions for the competence-oriented design of teaching and learning environments. It has to be examined whether the development and promotion of certain competencies, skills and abilities is influenced by the use of ChatGPT or comparable tools and should be adapted if necessary. One level higher still, the question arises as to whether there are now new or changed competencies in the respective area that students need for the constantly changing world of life and work.

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